

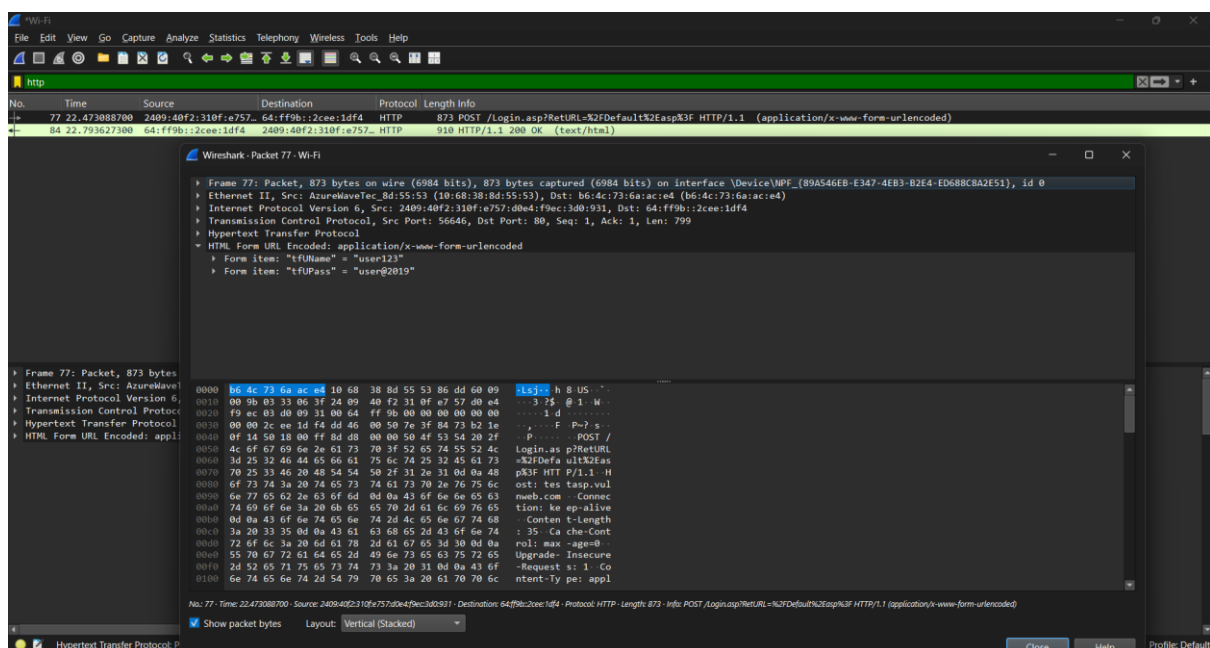
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Importance of Cryptography: Using Wireshark, investigate the transmission of dummy login credentials over HTTP and HTTPS

This experiment was to understand how login credentials are transmitted over HTTP and HTTPS. I used Wireshark to capture the network traffic and observe the difference between HTTP connection and HTTPS connection.

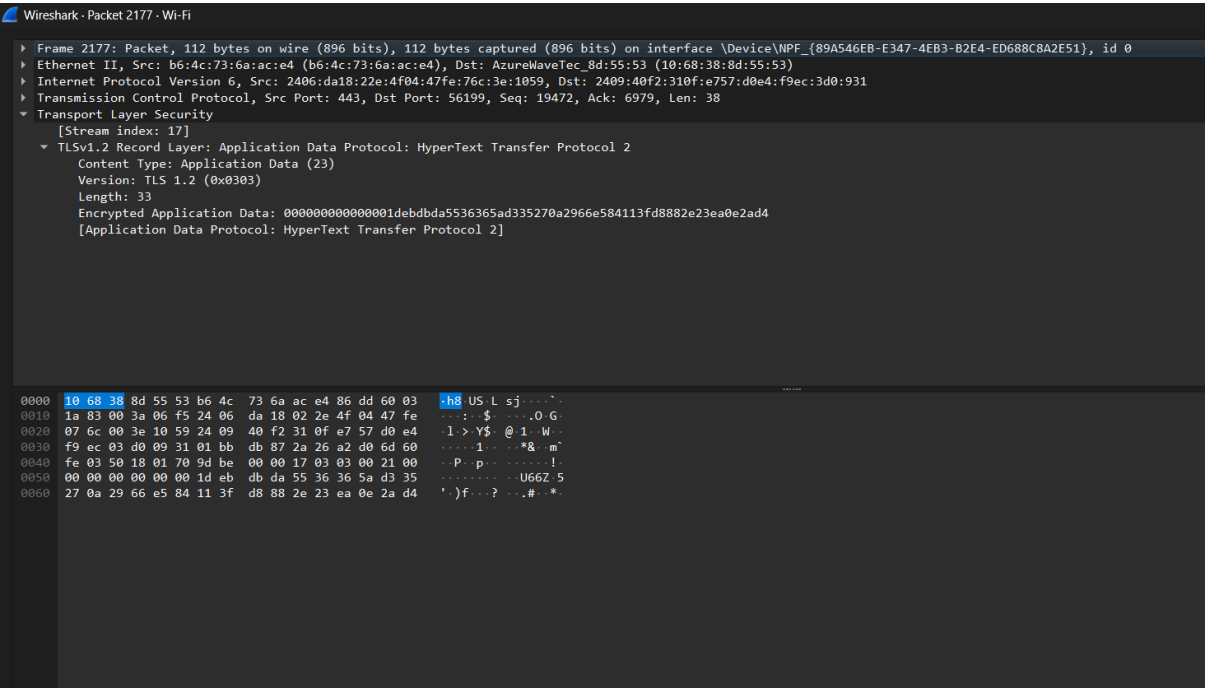
In this experiment we will open an HTTP website which contains a login page and allows us to enter Login credentials username and password and start Wireshark to capture the packets. The same is done on HTTPS website also and the packets are captured to observe the difference.

HTTP Traffic Analysis Observations



When we log into an unsecured website (HTTP), your information is not hidden or scrambled. Anyone who intercepts the network traffic can clearly see exactly what we typed-like the username "user123" and password "user@2019" as normal, readable text. Because this data is completely unprotected, someone snooping on the network can easily steal your credentials. This is why it is extremely dangerous to enter passwords or personal details on websites that aren't properly secured.

HTTPS Traffic Analysis Observations



When we log into a secure website (HTTPS), our information is encrypted into an unreadable code before it leaves the device. If someone intercepts your network traffic, they won't be able to see the username or password at all.

While eavesdroppers can still see basic details - like where the information is coming from and where it is going - the actual private message is completely locked and protected. This built-in security makes it impossible for snoops to simply read your data, making secure websites essential for keeping your sensitive information safe.

Comparison

Feature	HTTP	HTTPS
Encryption	No	Yes, using TLS
Data Visibility	Data can be readable	Data is encrypted
Port	80	443
Security	Less secure	More secure
Credentials	Can be exposed	Protected
Confidentiality	Not provided	Provided

Role of encryption

The only change between the two runs is the scheme in the URL. The portal, the password, and the server are identical. Over HTTP the password reaches the network as the characters typed into the form, so a sniffer on the path reads it with no effort. Over HTTPS, TLS wraps the request. TLS contributes in three ways. It keeps the payload confidential, which is what hides the password. It protects integrity with a message authentication code, so any change to the ciphertext in transit is detected. And in a real deployment it authenticates the server through a certificate signed by a trusted authority, which prevents an attacker from posing as the site. This lab uses a self-signed certificate, so the client warns that it cannot vouch for the server's identity, yet the encryption still functions. That warning concerns trust, not the strength of the encryption. This gap is the reason a login page must run over HTTPS. Plain HTTP trusts every device between the browser and the server: the local network, a rogue Wi-Fi access point, an ISP, or a compromised router. Any one of them can read the password. HTTPS removes that trust from the network and places it in the keys instead, and the packet capture demonstrates the result. One run leaves a password that anyone on the network can copy. The other leaves a block of bytes that is useless without the session secret.

Conclusion

The capture makes the difference concrete. Over HTTP, `admin` and `supersecret123` are recovered directly from the captured POST packet, where Wireshark decoded the form fields sent across the network. Over HTTPS, the same login produces only TLSv1.3 records, and the Application Data payload is ciphertext. Encryption is what closes that gap, and it is the reason credentials must never travel over plain HTTP.